

represented on the circuit. Modern distributed systems partially solve this problem by locating remote multiplexing devices out in the field. The ultimate solution, however, is to be able to connect a reasonable number of sensors all located in the same area to the same cable. Although this will not happen overnight, you should be prepared to accept this tectonic shift in technology.

Know advanced process control

To get an edge over your competitors, it is always advisable to learn advanced process control with respect to the underlying theory, implementation studies, the benefits that its applications will bring and projections of future trends.

Initially, advanced process control meant any algorithm or strategy that deviated from the classical three-term PID controller. The advent of computers offered more convenient alternatives—feed forward control, multivariable control and optimal process control. Indeed, the proliferation of so-called advanced control methodologies can only be attributed to the advances made in the electronics industry, especially in the development of low-cost digital computational devices (circa 1970). Nowadays, advanced control is synonymous with the implementation of computer-based technologies.

Also, try to understand the impact of advanced process control on product yield, energy consumption, product quality, process safety, environmental emissions, etc. Usually, cost savings ranging from 2 to 6 per cent of the operating cost are observed with the implementation of advanced controls. These benefits are clearly significant and achieved by increasing process efficiency, hence allowing plants to be operated closer to their designed capacity.

You should regard advanced control as more than just the use of a multiprocessor computers or state-

of-the-art software environments. Neither does it refer to the singular use of sophisticated control algorithms. It describes a practice, which draws upon elements from many disciplines ranging from control engineering, signal processing, statistics, decision theory and artificial intelligence to hardware and software engineering.

The algorithms of control

Remember that control systems run according to the logical flow of the operating program. Knowledge of different controlling algorithms is always nice to have. Realise the actual logic behind the control systems. For example, statistical process control is a method for achieving quality control in manufacturing processes. It is a set of methods using statistical tools such as mean, variance and others, to detect whether the process observed is under control.

Model predictive control (MPC) is widely adopted in the process industry as an effective means to deal with large, multivariable constrained control problems. The main idea of MPC is to choose the control action by repeatedly solving online an optimal control problem. This aims at minimising a performance criterion over a future horizon, possibly subject to constraints on the manipulated inputs and outputs, where the future behaviour is computed according to a model of the plant.

PID-type controllers do not perform well when applied to systems with significant time-delay. Perhaps the best known technique for controlling systems with large time-delays is the Smith Predictor. It overcomes the debilitating problems of delayed feedback by using predicted future states of the output for control.

Currently, some commercial controllers have Smith Predictors as programmable blocks. There are, however, many other model-based control strategies that have dead-time compensation properties. These are useful for predictive constrained control. Predictive controllers can also be embedded

within an adaptive framework.

Most processes require monitoring of more than one variable. Controller-loop interaction exists in such a way that the action of one controller affects other loops in a multiloop system. Depending upon the inter-relationship of the process variables, tuning each loop for maximum performance may result in system instability when operating in a closed-loop mode. Loops that have single-input single-output (SISO) controllers may therefore not be suitable for these types of applications. These types of controllers are not designed to handle the effects of loop interactions. Try to understand how a model-based controller can be modified to accommodate multivariable systems.

Dynamic matrix control is also a popular model-based control algorithm. The process model is stored in a matrix of step or impulse response coefficients. This model is used in parallel with the online process in order to predict future output values based on the past inputs and current measurements.

The final bend

It is possible that your awareness about most of the aforementioned terms is from a notional perspective only. Don't worry. Utilise your industrial training or final-year project to your advantage. You can get a holistic overview of 'chip to ship' of a control loop only after completing a project.

Nearly all of Indian institutes are woefully lagging in terms of providing students with such opportunities. If you feel that you lag behind due to lack of practical exposure, a strategically chosen course may be the solution. I emphasise the word 'strategically' because that is what decides whether you will get the job passport or your money will go down the drain. So before choosing a course, judge the reputation of the institute, the certification system, the industry accreditation and also the course curriculum. ●

The author is a research analyst cum journalist at EBY